Julie Kallini

Ph.D. Student | NLP | Computational Linguistics

juliekallini.com
kallini@stanford.edu
juliekallini
jkallini

About Me

I am a second-year Ph.D. student in Computer Science at Stanford University, where I am part of the Natural Language Processing Group. Before graduate school, I spent nearly two years at Meta as a software engineer, applying machine learning to privacy challenges in advertisements. Prior to that, I earned my undergraduate degree from Princeton University, where I had the opportunity to conduct research in computational linguistics. I grew up in Los Angeles, California.

Research Interests

My work lies at the intersection of NLP, machine learning, and linguistics. I am particularly interested in tokenization and vocabulary, with the goal of improving language models' efficiency and fairness across languages. I am also interested in understanding what deep learning can reveal about linguistics and language acquisition.

Education

2023—Pres **Ph.D. in Computer Science** — Stanford University
- GPA: 4.21/4.00
- Advisors: Christopher Potts and Dan Jurafsky

2017—2021 **B.S.E. in Computer Science, Minor in Linguistics** — Princeton University
- GPA: 3.94/4.00 (*summa cum laude*)
- Advisor: Christiane Fellbaum

Papers

[1] **Julie Kallini**, Shikhar Murty, Christopher D. Manning, Christopher Potts, and Róbert Csordás. MrT5: Dynamic Token Merging for Efficient Byte-level Language Models. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*, 2025. In press.

[2] **Julie Kallini**, Isabel Papadimitriou, Richard Futrell, Kyle Mahowald, and Christopher Potts. Mission: Impossible Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, Bangkok, Thailand, August 2024. **Best Paper Award**.

[3] **Julie Kallini** and Christiane Fellbaum. What to Make of *make*? Sense Distinctions for Light Verbs. In *Global WordNet Conference 2023*, Donostia-San Sebastian, Spain, January 2023.

[4] **Julie Kallini** and Christiane Fellbaum. Computational Approaches for Understanding Semantic Constraints on Two-termed Coordination Structures. In *Text, Speech, and Dialogue*, Brno, Czech Republic, 2022.

[5] **Julie Kallini** and Christiane Fellbaum. A Corpus-based Syntactic Analysis of Two-termed Unlike Coordination. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, Punta Cana, Dominican Republic, November 2021.

Teaching Experience

2022–Pres **Instructor**, *Applied Machine Learning* — Uplimit (formerly CoRise)
Built scikit-learn projects and designed course materials for introductory ML course for working professionals; taught at weekly lectures and office hours.

2021 **Head TA**, *Machine Translation* — Princeton University
Designed coding exercises in Python and led weekly classes on NLP topics.

2020 **TA**, *Computer Networks* — Princeton University
Taught course material and ran weekly in-class discussion sessions.

2018–2019 **Lead Grader/CA**, *Algorithms and Data Structures* — Princeton University
Designed Java assignment rubrics; trained and supervised other graders.

Industry Experience

2021–2023 **Software Engineer**, Machine Learning for Ads Privacy — Meta
Built classifiers to detect sensitive topics in media on the Meta family of apps.

2020 **Software Engineer Intern**, Community Integrity — Meta
Designed data analysis and monitoring tools for content moderation.

2019 **Software Engineer Intern**, Android Development — Meta
Created an Android app that organizes and streamlines office hours queues.

Fellowships, Awards, and Honors

2024	**Best Paper Award** (for *Mission: Impossible Language Models*)	ACL 2024
2023	**NSF Graduate Research Fellowship**	Stanford University
2023	**School of Engineering Graduate Fellowship**	Stanford University
2023	**EDGE Fellowship**	Stanford University
2021	**Phi Beta Kappa**	Princeton University
2021	**Phillip Goldman '86 Senior Prize** (Top CS Academic Prize)	Princeton University
2021	**Outstanding CS Senior Thesis Prize**	Princeton University
2021	**Outstanding Student Teaching Award**	Princeton University
2019	**Tau Beta Pi** (Engineering Honor Society)	Princeton University